

Course Syllabus

CS330e: Elements of Software Engineering I

Unique Numbers: 50385, 50390

Instructor: Dr. Fares Fraij

Fall 2024

General Information

Course Description This is a course on software engineering using Python. It is strongly focused on using tools to improve the quality of software development, including automated builds with [make](#), [source control with git](#) and [GitLab](#), [unit testing with unittest](#), [code coverage with coverage](#), [continuous integration with GitLab CI](#), [automated documentation with pydoc](#), [Docker](#).

Prerequisites

- Computer Science 303E and 313E.

Lecture Time and Location

M W 09:30 a.m. - 11:00 a.m. UTC 3.132 (50385)
M W 11:00 a.m. - 12:30 p.m. UTC 3.112 (50390)

Textbooks The following textbook is required for reference:

[Extreme Programming Installed](#)

-

Instructor [Fares Fraij](#)

- Office hours: **M** 12:30 p.m. - 01:30 p.m. (at GDC 6.404)
Thu 11:00 a.m. - 12:00 p.m. (at GDC 6.404)
Note: No office hours in the first week of classes
- Contact: fares@cs.utexas.edu

Note: Most questions should be submitted to [ED Discussion](#) rather than by sending an email to the instructor.

Guidelines

Time

- This course includes **self learning**.
- Each student should continuously and constantly be reading and practicing.
- To do well in this class, you should expect to commit an average of **5 quality hours** per week outside of class as follows 1 hour per week of reading/studying and 4 hours per week of programming.
- You are expected to write up around **1,000 lines of code**.

Attendance and In-class participation

- **Attendance** and **participation** in class discussions are **required**.
- You are expected to show up and stay for the whole class.
- I'll present a piece of code and ask you about it.
- You **will not** be responsible for the **correct** answer.
- The goal of participation is to **increase** the level of **interaction** in the class and help you **learn**.
- If you provide **an incorrect answer**, it's an opportunity to delve into **the thought process** that leads to **the correct solution** and correct fundamental misunderstanding.

Materials

- The **syllabus**, the **schedule**, the **projects** and the **course material** of the course are available on-line and can be accessed through the **instructor's web page**.
- **Additional course material** will be made available through [Canvas](#).
- You are **responsible** for **knowing** about all the **material** distributed for this class, whether it is mentioned in the **syllabus**, posted on the **class website**, posted on **Canvas**, posted on **Ed Discussion**, or announced **during lectures**.

Reading

- There will be **extra readings** assigned before almost every class.
- Each student should **read** the assigned material **before** coming to the class. Some questions in the Quizzes may test your knowledge of the readings.

Regrading Policy

- The TAs will grade and post the grades of the projects/assignments within **one-week**.
- Ed Discussion posts will be made within 24 hours of the project/assignment grades being released, indicating which TA graded each project/assignment.
- After receiving your grades for **projects**, or **assignments**, you have **only one week** to contest your grades.
- To request a regrade, prepare a text file with the following information:
 - Your **name**, **UT EID**, and **unique course ID** (e.g. 51470)
 - The **project/assignment number** you want regraded or point out a specific problem associated with your grade for example scores were added incorrectly.
 - Why you think the grade was in error (broken down by the problem, if possible)

- Send this information to the TA who graded the question either by **email** or [Canvas](#) message.
 - If the **one week** time frame is **passed**, **no changes** in grades will be made.
- Note:* that regrades will not be considered for “judgment calls” (e.g. you think a mistake should only be -5 points, but it was marked as -7).

Academic Honesty Students who violate University rules on academic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. Since such dishonesty harms the individual, all students, and the integrity of the University, policies on academic dishonesty will be strictly enforced.

Quizzes will be conducted using Canvas. You must take the quiz in class. Sending and/or receiving the access code, the questions, or the answers to anyone is a violation of the honor code of the University.

Programming assignments will be compared using [Moss](#). You **may** share **design ideas** with your fellow students. You **may not** share **code** in **any way** with your fellow students.

The penalty for academic dishonesty will be a course grade of **F** and a referral of the case to the [Dean of Students](#). Further penalties, including suspension or expulsion from the university, may be imposed by that office.

To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at: <http://deanofstudents.utexas.edu/conduct>.

Incompletes To be eligible for an **incomplete**, you must have a letter grade of **C** or better and a written, verifiable excuse for missing the last test. This is a **necessary** but **insufficient** condition for receiving an incomplete.

Accommodations for Religious Holidays By UT Austin policy, you must notify me of your pending absence at least **14 days** prior to the date of observance of a **religious holy** day. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Disabilities Students with disabilities may request an [official letter](#) outlining authorized accommodations from the Division of Diversity and Community Engagement, Services for Students with Disabilities, you can either call 512-471-6259 or visit the following website [Services for Students with Disabilities](#)

You must provide this letter to me by the **12th** class day. **Projects, quizzes and assignments deadlines** will **not** be **extended** for special-needs students.

Assessment:

Projects

- There will be 2 projects.
- Both projects may be done in pairs with pair programming. One solution will be turned in for the pair.
- Both students must vouch for the fact that they worked on the project together for more than 75% of the time.
- Projects must run in the CS Labs and on GitLab CI.
- For every day that a project is late, one-fifth of the points will be lost, so that in five days it will be worth zero. Corrupt or incomplete submissions will count as one day late. You must e-mail the grader when submitting late or re-submitted projects.
- Projects will first be graded without regard to penalties or bonuses. After that grade is determined, penalties will be subtracted and bonuses will be added, up to the max score.
- The solution to the project will not be provided. The grader will grade, comment, return, and post the grade of the project within one week. The grade can only be disputed within the following week. All grade disputes must be made in writing by e-mail to the grader with supporting evidence or arguments.
- If bonus pts are provided, they will not increase the total score beyond the max score.

Quizzes

- There will be regular quizzes throughout the semester.
- Quizzes will be and must be taken in-class. We may have more than one quiz per class. The first one is at the beginning of the class.
- Each quiz may cover the material of the previous class(es) and/or the current class and the assigned readings.
- There will be no makeup quizzes for any reason.
- The lowest 3 quiz scores will be dropped.
- These drops will take care of any reasons you may have in missing quizzes.
- If you take the quiz and leave or don't answer when calling your name for participation, I'll give you a zero for the quiz.

Assignments

- There will be several assignments throughout the semester.
- You must have a computer machine. If you don't, you must inform the instructor immediately.
- The assignments will be published on [Canvas](#). It's your responsibility to consistently check [Canvas](#).
- The outcomes of the tasks must be submitted using [Canvas](#)' assignments page.
- Assignments must be submitted in the specified file format.
- Assignments must run in the CS Labs.
- After submitting your assignment, make sure that your submission has been properly uploaded.
- For the first day that an assignment is late, 10 percent of the points will be lost, for the second day, an extra 10 percent of the points will be lost, in three days it will be worth zero.

Tests

- There will be 2 tests.
- There will be no final.
- The solution to the test will not be provided. The grader will grade and post the grade of the test within one week. The grade can only be disputed within the following week. All grade disputes must be made in writing by e-mail to the grader with supporting evidence or arguments.
- If you need more information regarding a specific question in your test, you can contact the grader who graded that question during his/her office hours.

**Extra
Credit**

A student can earn an additional 5 pt (0.5%) for filling out the class evaluation at the end of the term.

Tentative Course Schedule (subject to change)

<i>Week</i>	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>
1. Aug 24 makefile	First class --- Syllabus --- Introduction --- notes_1.txt		Hello.py Assertions.py notes_2.txt --- advice_make_classwork.txt make utility tutorial		
2. Aug 31 XP: Ch. 1-3	Labor Day - No Class		make utility (revisited) classwork		

<i>Week</i>	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>
3. Sep 7 Git Guide --- XP: Ch. 4-6	Git and GitLab tutorial classwork --- Assignment #1		examples_python_Docker.txt --- Project #1 --- phase_I.txt		
4. Sep 14 Paper #1: Pair Programming --- XP: Ch. 7-10	Project #1 --- unit tests tutorial classwork --- UnitTests1.py UnitTests2.py UnitTests3.py --- Assignment #2 --- notes_3.txt		Coverage1.py Coverage2.py Coverage3.py --- Project #1 --- phase_II.txt		
5. Sep 21 Paper #2: Agile	Project #1 --- phase_III.txt		Factorial.py FactorialT.py ---		

Development -- XP: Ch. 11-13	--- Assignment #3		notes_6_prime.docx --- IsPrime1.py IsPrime1T.py --- IsPrime2.py IsPrime2T.py --- notes_4.txt		
6. Sep 28 Paper #3: Collegiate Happiness --- XP: Ch. 14-16	CS330e_object_model Types.py --- Exceptions.py --- notes_5.txt		Operators.py --- Variables.py --- notes_6.txt		

<i>Week</i>	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>
7. Oct 5 Paper #4: Team Collaboration	Reduce.py ReduceT.py --- Iteration.py --- Yield.py --- notes_7.txt		RMSE.py RMSET.py --- Map.py MapT.py --- notes_8.txt	Project #1	
8. Oct 12 Paper #5: The Joel Test --- XP: Ch. 17-19	Comprehensions.py --- Iterables.py --- FunctionKeywords.py FunctionDefaults.py FunctionUnpacking.py FunctionTuple.py FunctionDict.py --- notes_9.txt		Test 1 (16 Oct) --- Assignment #4		

9. Oct 19 Paper #6: Continuous Integration --- XP: Ch. 20-21	Project #2 --- Cycle.py CycleT.py --- Count.py CountT.py --- Repeat.py RepeatT.py		Chain.py ChainT.py --- CCount.py CCountT.py --- CRepeat.py CRepeatT.py		
10. Oct 26 Paper #7: What Should We Teach --- XP: Ch. 22-23	CCycle.py CCycleT.py --- RangeIterator.py RangeIteratorT.py --- Range.py RangeT.py -- notes_10.txt		closures --- Lambdas.py --- notes_11.txt --- Assignment #5		

<i>Week</i>	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>
11. Nov 2 Paper #8: A Woman's Place	Decorators - revisited Decorators.py DecoratorsT.py --- Functions.py --- notes_12.txt		Select.py SelectT.py --- notes_13.txt		
12. Nov 9 Paper #9: Guerilla Interviewing	Project.py ProjectT.py --- CrossJoin.py CrossJoinT.py --- ThetaJoin.py ThetaJoinT.py		NaturalJoin.py NaturalJoinT.py --- JSON --- notes_15.txt	Project #2	

	--- notes_14.txt --- Assignment #6				
13. Nov 16 Paper #10: Terrible Female Engineers	Person.json Person.schema.json Person.schemaRestricted.json --- MovieClasswork.json MovieClasswork.schema.json --- Bookstore.json Bookstore.schema.json --- dict_file.py		XML --- BusinessCard.xml BusinessCard1.xml BusinessCardDTD.dtd		
14. Nov 23	Thanksgiving: No Class		Thanksgiving: No Class		
15. Nov 30 Paper #11: Résumé Read	People.xml People1.xml --- Bookstore1.xml Bookstore1.dtd.xml --- RegExps.py	Assignment #7	Regular Expressions (revisited) --- API api.py --- mypy_getting_started Mypy.py --- StdDev.py StdDevT.py --- Optional readings: Cache.py Copy.py		

<i>Week</i>	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>
16. Dec 7	Test 2 (9 Dec)				

